TED ÜNİVERSİTESİ

Computer Engineering Department

CMPE 382: Operating Systems

Introduction and Overview

Instructor and TA

- Instructors:
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  - Room: F318,
  - Office Hours: Mo 10:00-11:00,
- TA
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  - Room: B343

Operating Systems

Course Catalog Description	• Introduction to operating systems. Historical advances. • Operating system structures. The shell. System calls. • Multi-processing. Multi-threading. • Synchronization. Deadlocks handling. • CPU scheduling. • Memory managements. Storage and file systems. Input/output.
Course Learning Outcomes (LO)	**Upon successful completion of this course, a student will be able to** 1. Recognize the organization of the computer systems. 2. Describe major components of the operating systems. 3. Implement multi-processing 4. Develop process concepts in terms of threading, scheduling, synchronization and deadlocks. 5. Manage computer memory. 6.Implement file and I/O subsystems

Study Material

Text Book:

- Modern Operating Systems, by Andrew S. Tannenbaum, Fourth Edition, Wiley.
- Operating Systems Concepts by Silberschatz, Galvin and Gagne. 10th Edition, Wiley.

Homework:

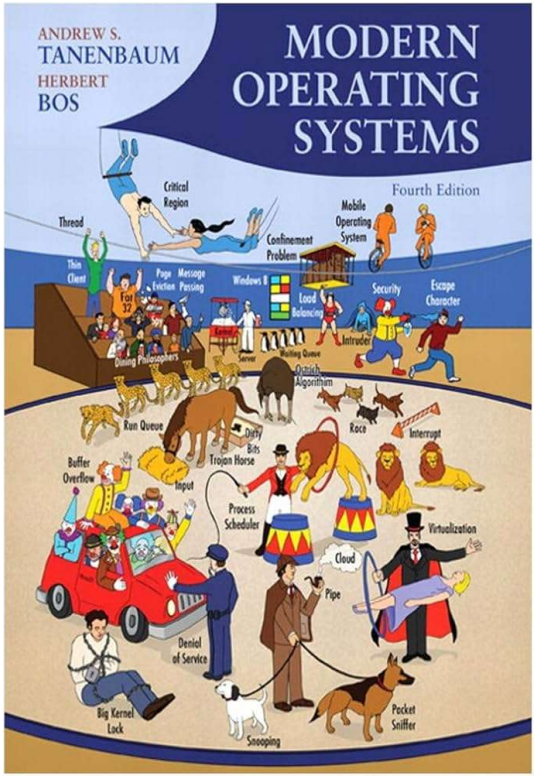
- Linux (or Unix), C Programing language

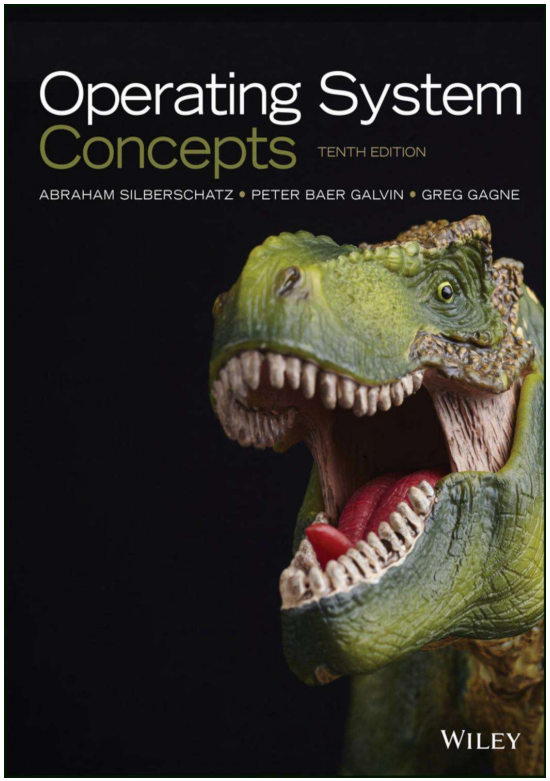
Announcements on LMS

- https://lms.tedu.edu.tr/

Attendance on EduKare App

- https://edukare.com.tr/





Assessment

Assessment	MT1	MT2	HW	Final	Total
Points	25%	25%	15%	35%	100

- HW
  - There are 3 HW (uploads).
  - No makeup for HW

Attendance Rules

Attendance is mandatory

- Students have to attend at least 18 Course Hours (17 hours of attendance will get FX grade)
- Students attended 28 Course Hours or more will be included in curving (from 27 to 18 hours of attendance will get catalogue grades)
- Check your attendance status from EduKare App, contact to your instructor immediately if there is any mistake

Make-up rules

Final and Midterm Exams:

- Only for students with valid medical report submitted to dean office according to the regulation.

Regrading

- You may request review of any graded item **only during the announced «Objection Period»**

Week	Dates	Topics	Learning Outcome (LO)	Textbook Reading	Assignments / Exams
W1	9 Feb – 13 Feb	Introduction to operating systems		Ch1	
W2	16 Feb – 20 Feb	Operating system structures and hardware properties		Ch1	
W3	23 Feb – 27 Feb	Process concept		Ch2.1, 2.2	
W4	2 Mar – 6 Mar	Multi-processes		Ch2.2, 2.3	
W5	9 Mar – 13 Mar	Thread concept and multi threading		Ch2.3	HW1
W6	16 Mar – 20 Mar	Inter Process Communication-IPC		Ch2.3	
W7	23 Mar – 27 Mar	Synchronization		Ch2.3	Midterm 1
W8	30 Mar – 3 Apr	CPU Scheduling			
W9	6 Apr – 10 Apr	Deadlocks in OSs		Ch2.4	HW2
W10	13 Apr – 17 Apr	Basics of Memory management		Ch6	
W11	20 Apr – 24 Apr	Virtual memory management 1		Ch3.1, 3.2	Midterm 2
W12	27 Apr – 1 May	Virtual memory management 2		Ch3.3	
W13	4 May – 8 May	File systems		Ch 3.4, 3.7	HW3
W14	11 May – 15 May	I/O and Disk subsystem, Security		Ch 4	
W14	18 May – 22 May	Compensation for vacations during the term		Ch 5, 9	

The Aim of the course

- WILL NOT TEACH YOU HOW TO USE AN OPERATING SYSTEM.
- WILL HELP YOU TO ACHIEVE AN UNDERSTANDING OF HOW AN OPERATING SYSTEM WORKS.

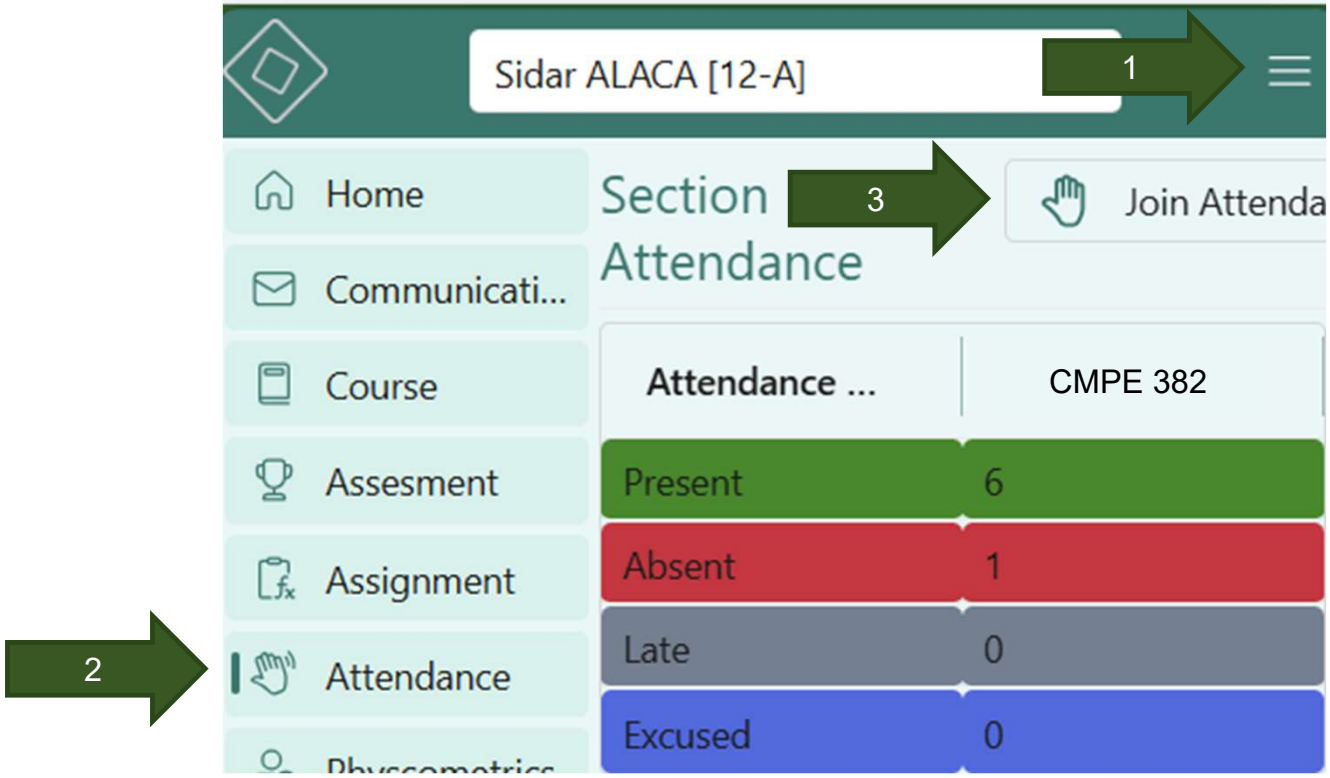
Keep in Mind

- Slides are not the main course materials
- They are guidelines to follow
- Use your main textbook for complete coverage

History: Vacuum Tube computers
https://www.youtube.com/watch?v=WnNm_uJYWhA

History of programming
https://youtu.be/kKJxzay85Vk?si=oAfsEkSSMrJJZvGU

Attendance Time



- You can leave if you can join attendance successfully